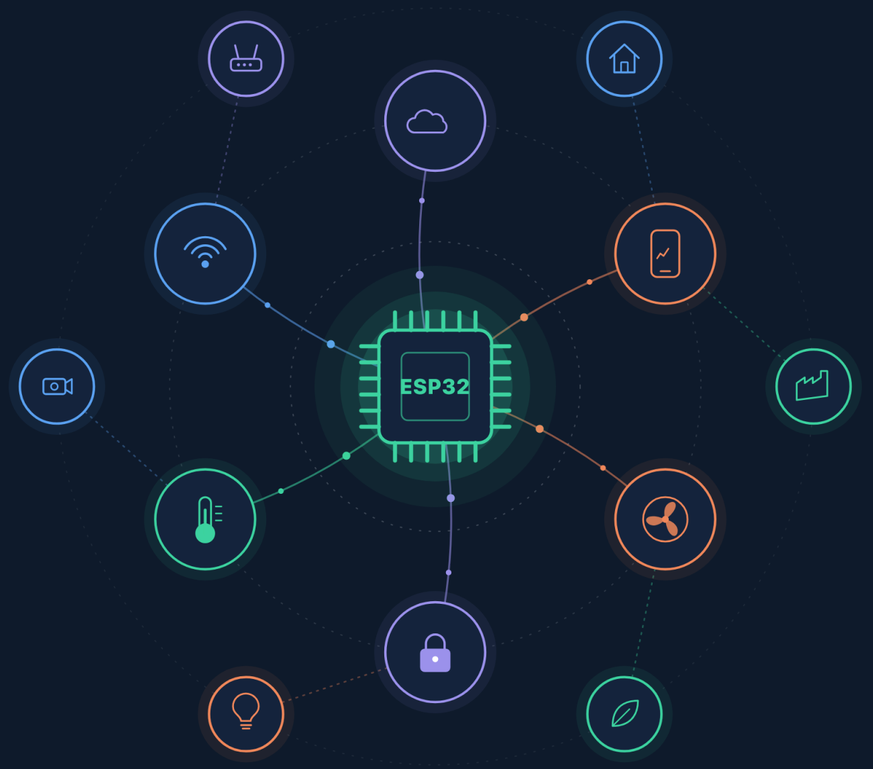
MODULE 3 · 45 MINUTES

Sensors, Actuators, and Reading Data

How a device feels the air in a tower shelter, switches on extra cooling, and turns signals into numbers.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
Lecture

02
How it's built
Lecture

03
Sensors and actuators
You are here

04
Getting online
Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05
Cloud and dashboards
Lab 4

06
Automation
Lab 5

07
Security
Group activity

08
Real world + pitch
Lab 7

BY THE END OF THIS MODULE

You will be able to...

01 **Name** the sensors at a tower site and in a data hall

02 **Explain** how a relay lets a small device switch big equipment

03 **Tell** analog from digital, and how a signal becomes a number

04 **Build** a shelter sensor in the Wokwi simulator

05 **Weigh** the cost of a sensor against a site visit or an outage

THE BIG PICTURE

The journey of one reading

Real world

The air in the shelter is 24 °C.

Sensor

Turns heat into an electrical signal.

Pin

The microcontroller reads the signal.

Your code

Converts the signal into a number.

24.0 °C

Ready to show, or to send to the NOC.

Actions go the other way: code decides → a pin sends a signal → an actuator acts.

Part A

How devices feel the world

What we sense at a tower site and in a data hall.

REMOTE SITE MONITORING

Sensors at a remote tower site



Fuel level

How much diesel is left for the generator.



Battery voltage

Are the backup batteries charged and healthy?



Door contact

Is the shelter door open or closed?



Temp & humidity

The air in the equipment shelter. Too hot? Too damp?

Business lens: fuel that drops while the generator is **off** is suspicious. Possible theft.

OUR SECOND CASE

Sensors in our data centres and exchanges

Temp & humidity

The air going into each rack. Too hot? Too damp?

Water-leak cable

Lies on the floor. Says “wet” or “dry”.

Smoke detector

Early fire warning.

Power meter

A current clamp: how much power a rack uses.

The readings go to **DCIM** or the **BMS**: the software your operations team already uses.

THE BUSINESS LENS

A sensor versus an outage

No. 1

cause of serious outages: **power failures**

1 in 5

operators: last major outage cost **over US$1 million**

Ask: what does a sensor cost? A site visit? One missed warning? → **Canvas box 1**

Source: Uptime Institute, Annual Outage Analysis 2026

Part B

Actuators: how devices act

Light, sound, switching and movement.

FOUR COMMON ACTUATORS

From a signal to an action

LED

A status light: green for OK, red for a problem.

Buzzer

An alarm beep, to get attention.

Relay

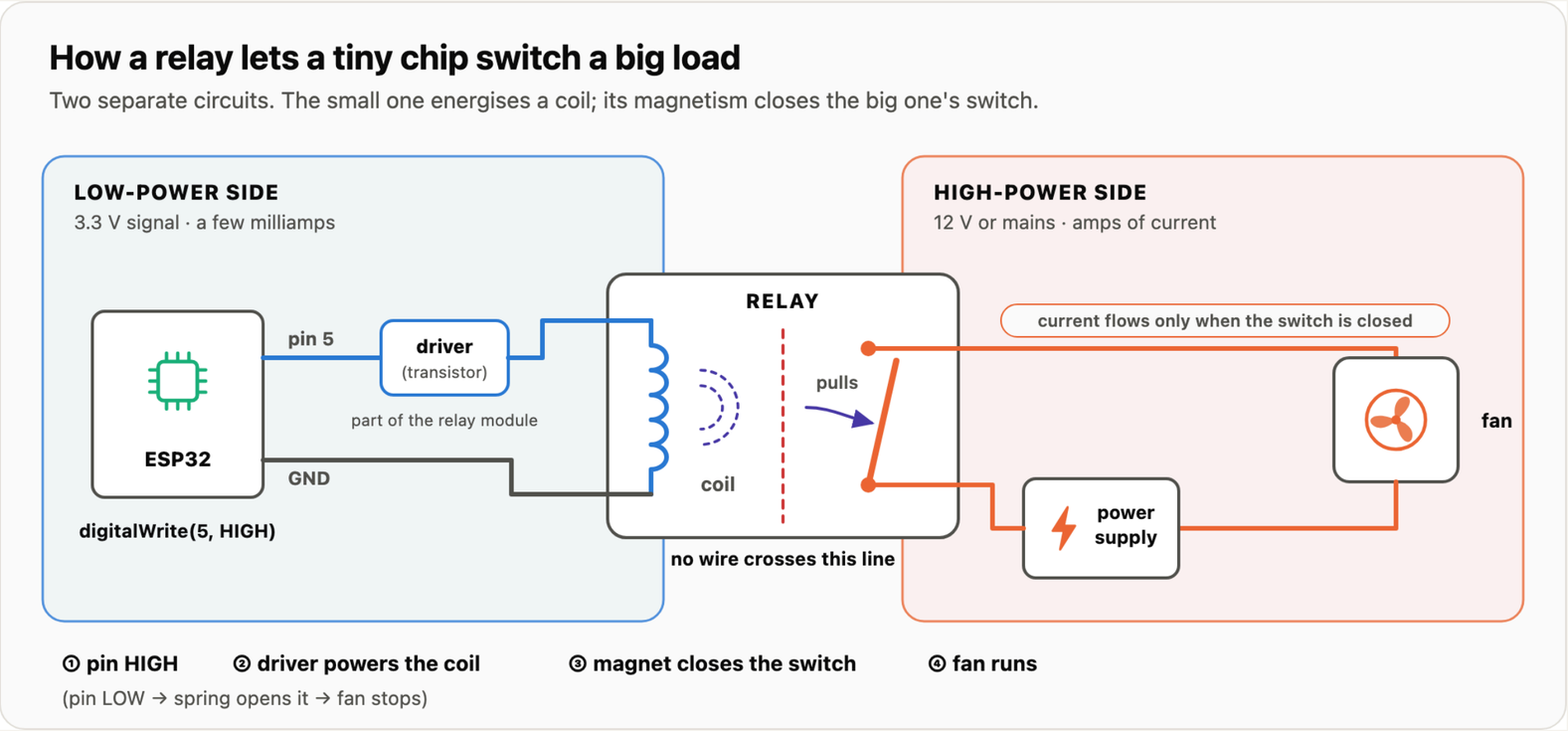
Switches big equipment, like extra cooling.

Motor

Movement, like opening a vent. Needs a driver.

HOW A RELAY WORKS

A small signal switches a big circuit



A pin gives only a **small** amount of power

The coil’s **magnetism** closes the switch

The two circuits stay **separate**

Part C

Analog vs digital signals

The most important technical idea today — and simpler than it sounds.

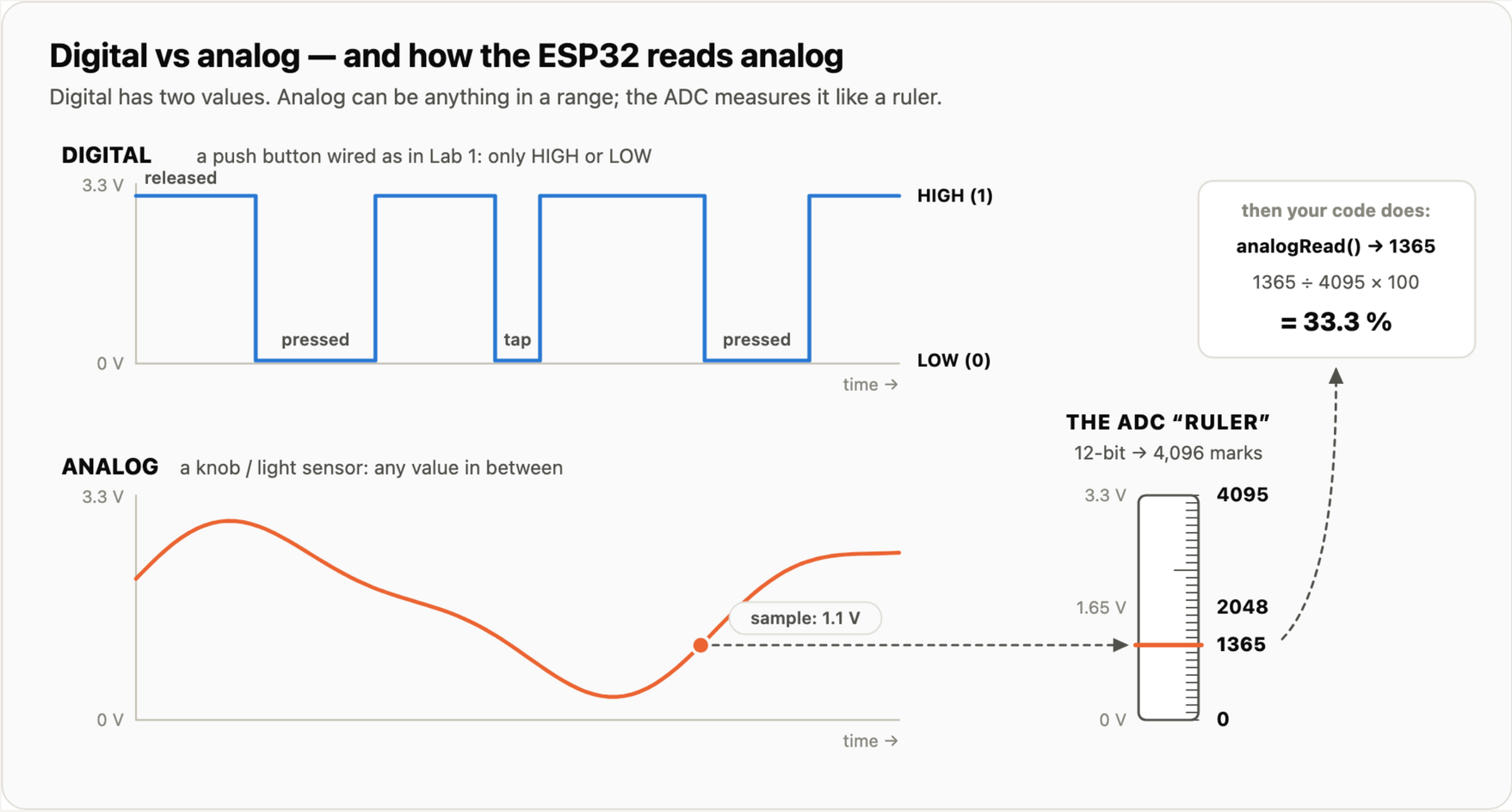
TWO KINDS OF SIGNAL

Digital and analog, side by side

	Digital	Analog
Values	Two: on/off, HIGH/LOW, 1/0	Any value in a range
Like a…	Light switch	Volume knob
Examples	Door contact, smoke alarm, button	Battery voltage, fuel level, dial
Read with	digitalRead()	analogRead()
It answers	“Is it happening?”	“How much?”

SEEING THE DIFFERENCE

From a voltage to a number



THE ESP32’S 12-BIT ADC

analogRead() gives 0 to 4095

0

at 0 V (nothing)

≈2048

at about 1.65 V (half-way)

4095

at about 3.3 V (full)

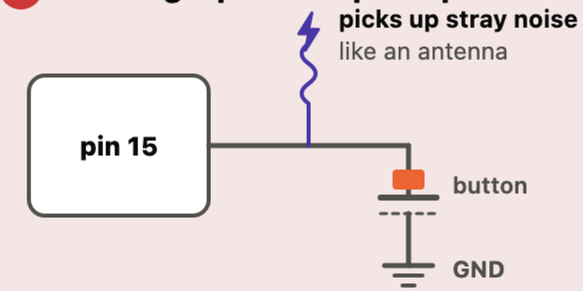
Read a raw number, then **convert** it: (raw ÷ 4095) × 100 = percent. With Wi-Fi, use **ADC1 pins (GPIO 32–39)**.

Don’t leave inputs floating

Why buttons need a pull-up resistor

An unconnected input “floats” and reads random noise. A pull-up holds it steady until pressed.

Floating input — no pull-up

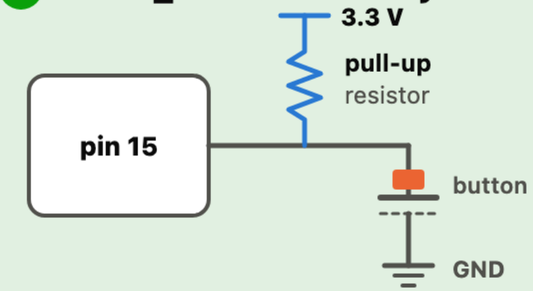


What digitalRead() returns, reading after reading:

H L H H L H L L H L H H L H

Random — the code sees presses that never happened.

INPUT_PULLUP — steady and reliable



What digitalRead() returns, reading after reading:

H H H H H L L L L H H H H H

button held down

HIGH at rest, LOW only while pressed.

Code: pinMode(15, INPUT_PULLUP);

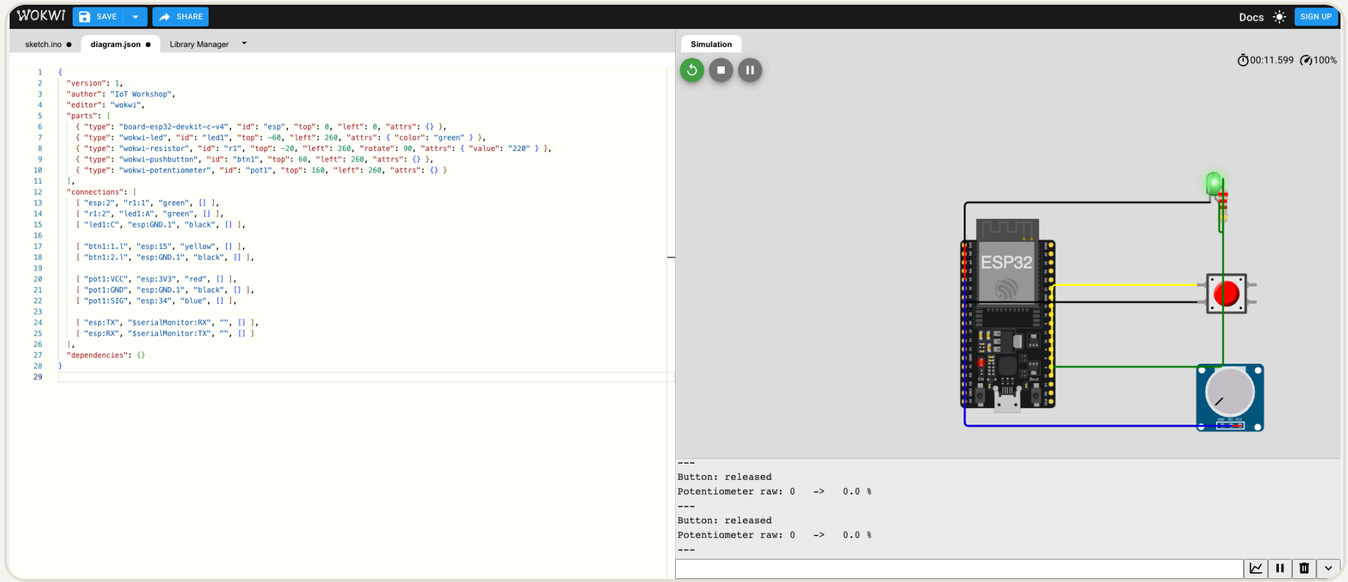
PAUSE AND CHECK

Your tower's shelter door alarm says "open" at random, but nobody is there. Why, and what's the fix?

Think about the picture we just saw.

LIVE DEMO · LAB 1 · 3 MINUTES

Watch: digital vs analog in Wokwi



1. I paste the code and circuit, press **Play**
2. The LED blinks: a **digital output**
3. I hold the button: **PRESSED**, a digital input
4. I turn the dial: **0 to 4095**, an analog input
5. Try it yourself later with the guide

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

IN TELCO OPERATIONS

What sensors change at a tower site

	Today	With IoT	The benefit
Hot shelter	Found at the next visit, or when equipment fails	The NOC gets an alert; extra cooling switches on	Faster fixes, fewer failures
Low battery	Found when the site goes down in a power cut	Battery voltage is watched; alert before it runs low	Fewer outages
Open door	Nobody knows until the next visit	The NOC gets a door alert at once	Less theft, safer sites

All three are **remote site monitoring**: fewer trips just to look.

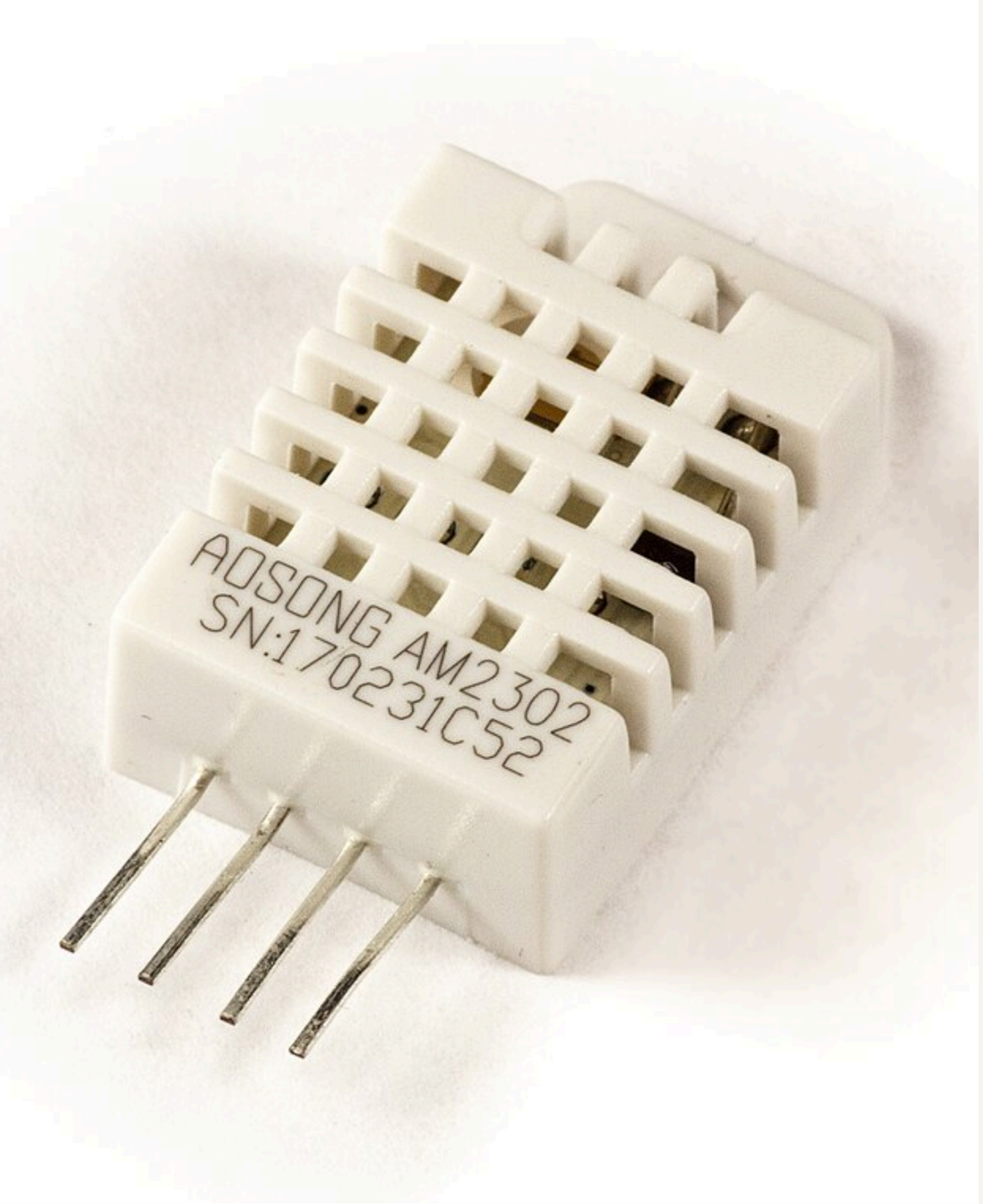


Photo: Ubahnverleih, CC0, via Wikimedia Commons

THE SENSOR FOR LAB 2

The DHT22

- Temperature **and** humidity in one small part
- A fresh reading only every **2 seconds**
- Accurate to about **±0.5 °C**
- Today: your **shelter** sensor



Photo: pslim, CC BY 2.0, via Wikimedia Commons

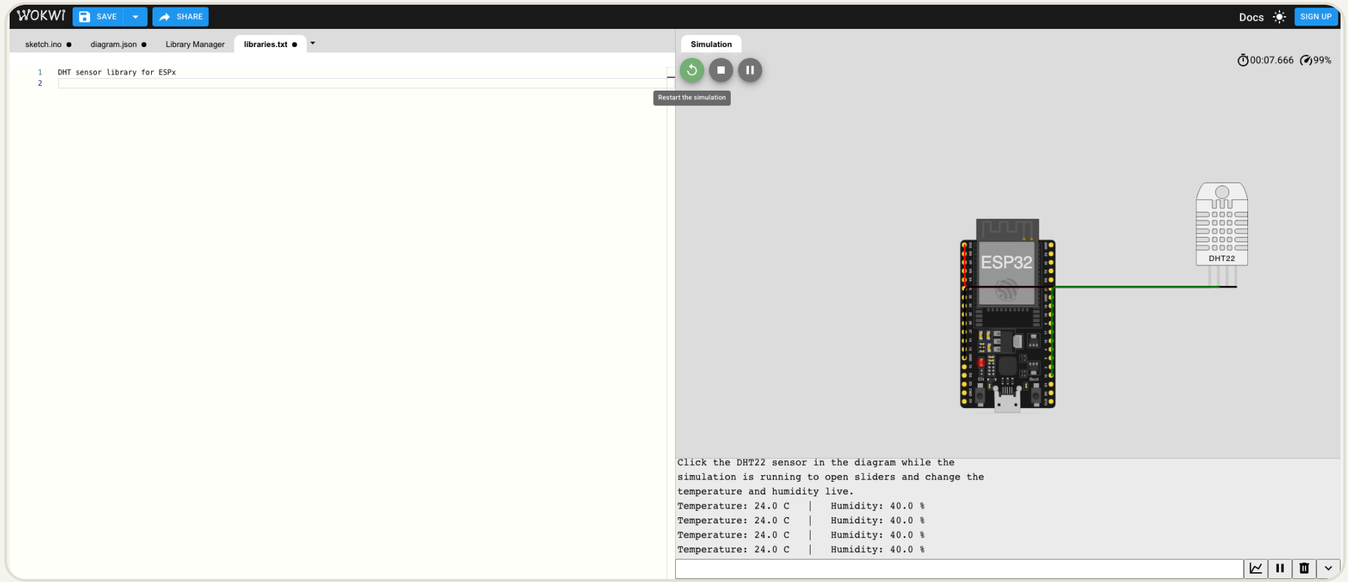
WHERE THE SENSOR GOES

Inside the equipment shelter

- A **shelter** holds the radios and batteries
- Nobody is there, so the **sensor** tells the NOC
- Too hot? Extra cooling **on at 28 °C**, off at 26 °C
- Which **five readings** from every site? **You decide**

LAB 2 · HANDS-ON · 20 MINUTES

Build the shelter sensor



1. Paste **sketch.ino** and **diagram.json**
2. Add **libraries.txt** with the DHT library name
3. Press **Play**: you see 24.0 °C and 40 %
4. Click the DHT22 and drag its sliders
5. Decision owner: which **five readings**?

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- **Sensors** turn heat, fuel level or an open door into signals
- **Actuators** act; a **relay** lets a small pin switch big equipment
- **Digital**: “Is it happening?” **Analog**: “How much?”
- The **ADC** gives 0–4095; the code **converts** it to real units
- Weigh the **sensor’s cost** against a **site visit** or an **outage**

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 Name one digital sensor and one analog sensor at a tower site.

Q2 A fuel sensor’s analogRead() returns 1024. About how full is the tank?

Q3 Why can’t you connect the shelter’s cooling fan straight to an ESP32 pin?

Q4 The shelter’s DHT22 is read ten times a second. Why is that a problem?

UP NEXT · MODULE 4

Getting Online

How the data travels: Wi-Fi, cellular IoT and MQTT — and your shelter readings sent across the internet in Lab 3.

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Book, labs and slides: github.com/datatweets/intro-to-iot-workshop